Representation Theory

Labix

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Abstract

Contents

1 Basic Definitions of Representations

1.1 Representation of an Algebra

Definition 1.1.1 (Representation of an Algebra)
Let R be a commutative ring. Let A be an R-algebra. A representation of A is a left R-module M and a homomorphism of R-algebras

$$\rho : A \to \text{End}_R(M)$$

Let A, M be sets. Recall that there is a natural bijection

$$\text{Hom}_{\textbf{Sets}}(A \times M, M) \cong \text{Hom}_{\textbf{Sets}}(A, \text{Hom}_{\textbf{Sets}}(M, M))$$

given by curling. A representation of an R-algebra A is an element in

$$\text{Hom}_{\textbf{Alg}_R}(A, \text{End}_R(M)) \subseteq \text{Hom}_{\textbf{Sets}}(A, \text{Hom}_{\textbf{Sets}}(M, M))$$

Therefore we may ask what kinds of maps of the form $A \times M \to M$ give a representation of A.

Proposition 1.1.2
Let R be a commutative ring. Let A be an R-algebra. Let M be an R-module. Let $f : A \times M \to M$ be a function. Then the map $a \mapsto f(a, -)$ defines a representation of A if and only if the following are true.

- f is an R-module homomorphism.
- f defines an A-module structure on M.

The A-module structure can also be seen in the representation $\rho : A \to \text{End}_R(M)$. It is given by $a \cdot m = \rho(a)(m)$. In particular, not every A-module M gives a representation; the action of A on M must be bilinear in the scalars R. Therefore these modules are often called R-linear A-modules.

However, if $R = k$ is a field and the structure map $k \to A$ is non-zero, then $k \to A$ is necessarily injective so that A contains k as a subring. Then the k-linearity of $f : A \times M \to M$ is null so that A-modules are the same as representations of A over k.

Since representations of an algebra A are A-modules, it follows that the following notions in Rings and Modules apply to representations.

- Subrepresentations.
- Quotient Representations.
- Sum of Representations.

Definition 1.1.3 (Types of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M be a representation of A.

- We say that M is irreducible if it is simple as an A-module.
- We say that M is semisimple if it is semisimple as an A-module.
- We say that M is indecomposable if it is not completely reducible.

Example 1.1.4
Let k be a field. Define a representation $\rho : k[x] \to \text{End}_k(k^2) \cong M_2(k)$ by $x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then extended it R-linearly and respecting products of x. Then ρ is indecomposable but reducible.

Proof
Suppose for a contradiction that ρ is decomposable. Then there exists 1-dimensional linear subspaces L_1, L_2 of k^2 such that ρ_{L_1} and ρ_{L_2} are representations. Suppose that $L_1 = k\langle v_1 \rangle$ and $L_2 = k\langle v_2 \rangle$. Then $x \cdot v_1 \in L_1$ and $x \cdot v_2 \in L_2$ implies that v_1 and v_2 are linearly independent

eigenvectors of $\rho(x)$. But by an easy computation, we see that the only eigenvalue of $\rho(x)$ is 1 and its eigenspace has dimension 1. Hence this is a contradiction.

From the above, we see that the eigenspace of $\rho(x)$ is a subrepresentation of ρ. Hence ρ is reducible. ■

Definition 1.1.5 (The Regular Representation)
Let k be a field. Let A be a k-algebra. Define the regular representation $\text{reg} : A \to \text{End}_k(A)$ of A by
$$\text{reg}(a)(b) = ab$$

Definition 1.1.6 (Composition Factors)
Let R be a commutative ring. Let A be an R-algebra. Let M be a semisimple representation of A. Suppose that the irreducible decomposition of M is given by
$$M \cong \bigoplus_{i=1}^{n} M_i$$
Define the composition factors of M to be
$$\text{Comp}(M) = \{M_1, \dots, M_n\}$$

Definition 1.1.7 (Multiplicity of an Irreducible Submodule)
Let R be a commutative ring. Let A be an R-algebra. Let M be a semisimple representation of A. Let N be an irreducible A-module. Define the multiplicity of N in M to be
$$\text{mult}_N(M) = |\{M_i \in \text{Comp}(M) \mid M_i \cong N\}|$$

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M, N be representations of A. Let $f : M \to N$ be a function. We say that f is a homomorphism of representations if the following are true.
- f is an A-module homomorphism.
- f is R-linear.

Since M and N are A-modules, the definition of homomorphism implies that such a morphism must be an A-module homomorphism. Conversely, since not every A-module is an R-linear A-module, A-module homomorphisms are not necessarily morphisms between representations.

Definition 1.2.2 (Set of Homomorphisms of Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M, N be representations of A. Define the set of homomorphisms from M to N to be
$$\text{Hom}_{A_R}(M, N) = \left\{ f : M \to N \;\middle|\; \substack{f \text{ is a homomorphism} \\ \text{of representations from } M \text{ to } N} \right\} \subseteq \text{Hom}_A(M, N)$$

Definition 1.2.3 (Isomorphisms of Representations) Let R be a commutative ring. Let A be an R-algebra. Let M, N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. We say that ϕ is an isomorphism if there exists a homomorphism of representations $\psi : N \to M$ such that
$$\psi \circ \phi = \text{id}_M \quad \text{and} \quad \phi \circ \psi = \text{id}_N$$

Lemma 1.2.4
Let R be a commutative ring. Let A be an R-algebra. Let M, N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. Then ϕ is an isomorphism if and only if ϕ is an isomorphism of R-modules.

Theorem 1.2.5 (First Isomorphism Theorem for Representations)
Let R be a commutative ring. Let A be an R-algebra. Let M, N be R-modules. Let $\rho : A \to \text{End}_R(M)$ and $\tau : A \to \text{End}_R(N)$ be representations of A. Let $\phi : M \to N$ be a homomorphism of representations. Then the following are true.

- $\ker(\phi)$ is a subrepresentation of M.
- $\text{im}(\phi)$ is a subrepresentation of N.
- There is an isomorphism of representations
$$M/\ker(\phi) \cong \text{im}(\phi)$$
given by ϕ.

Proposition 1.2.6
Let R be a commutative ring. Let A be a finite R-algebra. Let $\rho : A \to \text{End}_R(M)$ be an irreducible representation of A. Then M is finitely generated.

Proof
Let $0 \neq m \in M$. Then $\rho(A)(m)$ is a submodule of M. Since M is irreducible, we must have $\rho(A)(m) = M$. Since A is finite over R, we must have $\rho(A)(m)$ is finitely generated over R. Hence M is finitely generated over R. ■

2 Finite Dimensional Representations of Algebra over Algebraically Closed Fields

2.1 Schur's Lemma and its Consequences

Recall from Rings and Modules that we have Schur's lemma: A homomorphism $M \to N$ of simple R-modules is either the 0 map or an isomorphism.

Lemma 2.1.1 (Schur's Lemma III)
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a finite dimensional k-vector space. Let $\rho : A \to \text{End}_k(V)$ be an irreducible representations of A. Let $\phi : V \to V$ be a homomorphism of representations. Then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.

Proof
Since ϕ is a homomorphism of simple A-modules, by Schur's lemma we know that $\phi = 0$ or ϕ is an isomorphism. If $\phi = 0$ then we are done. Otherwise, since k is algebraically closed, the endomorphism ϕ admits an eigenvalue $\lambda \in k$ and eigenvector $v \in V$. Consider the homomorphism of representations $\phi - \lambda \text{id}_V$. By Schur's lemma, this map is either 0 or an isomorphism. But it is not an isomorphism since $\phi(v) - \lambda v = 0$. Hence $\phi - \lambda \text{id}_V = 0$. So we are done. ■

Corollary 2.1.2
Let k be an algebraically closed field. Let A be a commutative k-algebra. Let V be a finite dimensional k-vector space. Let $\rho : A \to \text{End}_k(V)$ be an irreducible representations of A. Then $\dim(V) = 1$.

Proof
By Schur's lemma III, we know that $\rho(a) : V \to V$ is a scalar operator for all $a \in A$. This means that $\rho(a)(v) = \lambda_a v$ for some $\lambda_a \in k$ and all $v \in V$. For any fixed $v_0 \in V$, this implies that $k\langle v_0 \rangle$ is a subrepresentation of ρ. If $\dim(V) \geq 2$, then $k\langle v_0 \rangle$ is a non-trivial subrepresentation, contradicting the irreducible assumption. Hence $\dim(V) = 1$. ■

In other words, every finite dimensional representation of a commutative algebra over an algebraically closed field must be of dimension 1. And these are easy to classify.

Proposition 2.1.3
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be finite dimensional irreducible representations of A. Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proof
If V is not isomorphic to W, then by Schur's lemma we must have $\text{Hom}_{\mathbf{Rep}(A)}(V, W) \subseteq \text{Hom}_A(V, W) = \{0\}$. If $V = W$, then Schur's lemma III implies that id_V generates the k-vector space. ■

Proposition 2.1.4
Let k be an algebraically closed field. Let A be a k-algebra. Let V_i, W_j be finite dimensional irreducible representations of A for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we have

$$\dim_k\left(\text{Hom}_{A_k}\left(\bigoplus_{i=1}^n V_i, \bigoplus_{j=1}^m W_j\right)\right) = \sum_{i=1}^n \sum_{j=1}^m \dim_k(\text{Hom}_{A_k}(V_i, W_j))$$

Corollary 2.1.5
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be finite dimensional representations of A such that W is irreducible. Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \text{mult}_W(V)$$

2.2 Character Theory

Let k be an algebraically closed field. Recall that the trace of a k-linear map between finite dimensional vector spaces have a basis independent description.

Definition 2.2.1 (Characters of a Representation)
Let k be an algebraically closed field. Let A be a k-algebra. Let $\rho : A \to \text{End}_k(V)$ be a finite dimensional representation of A. Define the character of ρ to be function $\chi_V : A \to k$ given by

$$\chi_V(a) = \text{tr}(\rho(a))$$

Proposition 2.2.2
Let k be an algebraically closed field. Let A be a k-algebra. Let V, W be isomorphic finite dimensional representations of A. Then $\chi_V = \chi_W$.

Proposition 2.2.3
Let k be an an algebraically closed field. Let A be a k-algebra. Let $\rho : A \to \text{End}_k(V)$ be a finite dimensional representation of A. Then χ_V descends to a well defined function $\chi_V : \frac{A}{[A,A]} \to k$.

Lemma 2.2.4
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a finite dimensional representation of A. Let $W \leq V$ be a subrepresentation. Then we have

$$\chi_V = \chi_W + \chi_{V/W}$$

Definition 2.2.5 (Irreducible Characters)
Let k be an algebraically closed field. Let A be a k-algebra. Let V be a representation of A. We say that χ_V is irreducible if V is irreducible.

Proposition 2.2.6
Let k be an algebraically closed field. Let A be a k-algebra. Let $V_1, \dots, V_n$ be finite dimensional irreducible representations of A. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ are linearly independent in $\text{Hom}_k(A/[A, A], k)$.

Definition 2.2.7 (Characters of an Algebra)
Let k be an algebraically closed field. Let A be a k-algebra. Let $\phi : A \to k$ be a function. We say that ϕ is a character of A if there exists a finite dimensional representation V of A such that $\phi = \chi_V$.

2.3 The Semisimple Condition

Proposition 2.3.1
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A. Then there is a k-algebra isomorphism

$$A \cong \bigoplus_{i=1}^n \text{End}_k(V_i)$$

Proof
By the Artin Wedderburn theorem, there is a k-algebra isomorphism

$$A \cong \bigoplus_{i=1}^{p} M_{n_i}(D_i)$$

where each D_i is a division algebra over k. Since k is algerbaically closed, we have $D_i = k$. By a corollary of the Artin-Wedderburn theorem, the complete list of simple A-modules is given by the unique simple $M_{n_i}(k)$-submodule T_i of itself. Hence $n = p$. Moreover, by reordering the index, we have isomorphisms of $M_{n_i}(k)$-modules $V_i \cong T_i$ that are isomorphisms of A-modules (since the ring structure of the product of matrices is given by the product of the ring structures of the matrix ring, and the Artin-Wedderburn isomorphism is a ring isomorphism). Now $T_i \cong k^{n_i}$ as k-vector spaces. This gives an isomorphism $V_i \cong k^{n_i}$ of k-vector spaces and hence an isomorphism of rings $\text{End}_k(V_i) \cong M_{n_i}(k)$. Thus we have

$$A \cong \bigoplus_{i=1}^{p} M_{n_i}(k) \cong \bigoplus_{i=1}^{n} \text{End}_k(V_i)$$

■

Corollary 2.3.2
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A. Then we have

$$\dim_k(A) = \sum_{i=1}^{n} \dim_k(V_i)^2$$

Corollary 2.3.3
Let k be an algebraically closed field. Let A be a finite k-algebra. Let V be an irreducible representation of A. The we have

$$\text{mult}_V(A) = \dim_k(\text{Hom}_{A_k}(A, V)) = \dim_k(V)$$

Proof
Direct from Artin-Wedderburn theorem and its corollaries. ■

Proposition 2.3.4
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible finite dimensional representation of A. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is a basis for $\text{Hom}_k(A/[A, A], k)$.

Corollary 2.3.5
Let k be an algebraically closed field. Let A be a finite semisimple k-algebra. Let χ be a character of A. Then there exists a representation V of A such that $\chi = \chi_V$.

3 Representation Theory of Finite Groups

3.1 Structural Results

By the natural isomorphism

$$\mathrm{Hom}_{\mathbf{Alg}_k}(k[G], \mathrm{End}_k(V)) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, \mathrm{End}_k(V))$$

we see that representations of a group is the same as representation of the group ring.

Example 3.1.1
The following are representations of $D_8 = \langle r, s \mid r^4, s^2, rsrs\rangle$.

- $\rho : D_8 \to GL_2(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\rho : D_8 \to GL_n(\mathbb{C})$ defined by $\rho(r) = \rho(s) = I_n$.
- $\rho : D_8 \to \mathbb{C}^\times$ defined by $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$.
- $\rho : D_8 \to GL_4(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Definition 3.1.2 (Permutation Representations)
Let X be a set. Let G be a group acting on X. Let k be a field. Define the permutation representation $\rho : G \to \mathrm{GL}(k\langle X\rangle)$ of G on X over k by the formula

$$\rho(g)(x) = g \cdot x$$

and then extending it k-linearly.

Every group G has an action on itself by left multiplication. In this case the permutation representation is just the regular representation.

Example 3.1.3
The representation of $C_3 = \langle x \mid x^3\rangle$ on $\mathbb{F}_3^2$ given by

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not semisimple.

Let k be an algebraically closed field. Let G be a finite group. Suppose that $\mathrm{char}(k)$ does not divide $|G|$. Maschke's theorem which implies that $k[G]$ is semisimple is a powerful tool to classify irreducible representations of G because of the corollaries it gives from section 1.

- Since $|G|$ is finite, every representation of G is finite dimensional. Moreover, there are only finitely many non-isomorphic irreducible representations $V_1, \dots, V_n$.
- Combining the above corollary, we have a decomposition of the regular representation into irreducible representations
$$k[G] \cong \bigoplus_{i=1}^{n} \bigoplus_{j=1}^{\dim(V_i)} V_i$$
- This gives the formula
$$|G| = \sum_{i=1}^{n} \dim_k(V_i)^2$$

Example 3.1.4
Let G be a group of order 8. Let k be an algebraically closed field such that $\mathrm{char}(k) \neq 2$. Let

$V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$-modules. Then exactly one of the following is true (up to reordering of the modules).

- $n = 8$ and $\dim(V_i) = 1$.
- $n = 5$ and $\dim(V_1) = \cdots = \dim(V_4) = 1$ and $\dim(V_5) = 2$.

Similar to the case of representations, the character of a representation of a finite group $\chi : k[G] \to k$ is the same information as a function $\chi : G \to k$ simply be restriction and extension via scalars.

Proposition 3.1.5
Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let $\rho : G \to \text{End}_k(V)$ be a representation of G. Then the following are true.

- $\chi_V(1_G) = \dim(V)$.
- $\chi_V(hgh^{-1}) = \chi(g)$ for all $g, h \in G$.

The second item implies that χ descends to a function on conjugation classes $\chi : \text{Cl}(G) \to k$.

Proposition 3.1.6
Let G be a finite group. Let $\rho : G \to \text{End}_{\mathbb{C}}(V)$ be a complex representation of G. Let $g \in G$. Then the following are true.

- Let ω be a $|g|$th root of unit. Then we have
$$\chi(g) = \sum_{i=1}^{k} \omega^{j_i}$$
for some $j_i \in \mathbb{N}$ and $k \in \mathbb{N}$.
- We have $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Proof
Let $H = \langle g \rangle$. Then V is also a $\mathbb{C}[H]$-module. Let $V = \bigoplus_{i=1}^m V_i$ be the decomposition of V into irreducible $\mathbb{C}[H]$-submodules. Since H is cyclic, we must have $\dim(V_i) = 1$. ■

3.2 The Dual and Tensor Products of Representations

Definition 3.2.1 (The Dual Representation)
Let G be a finite group. Let $\rho : G \to \text{End}_{\mathbb{C}}(V)$ be a complex representation of G. Define the dual of ρ to be the representation $\rho^* : G \to \text{End}_{\mathbb{C}}(V^*)$ given by
$$\rho^*(g) = \rho(g^{-1})^T$$

Lemma 3.2.2
Let G be a finite group. Let $\rho : G \to \text{End}_{\mathbb{C}}(V)$ be a complex representation of G. Then we have $\rho^*(g)(f) = f \circ \rho(g)$ for $f \in V^*$.

Proposition 3.2.3
Let G be a finite group. Let V be a $\mathbb{C}[G]$-module. Then the following are true.

- We have $\chi_{V^*} = \overline{\chi_V}$.
- V is irreducible if and only if V^* is irreducible.

3.3 Orthogonality of Characters of Irreducible Representations

Let G be a finite group. Since $\mathbb{C}[G]$ is a semisimple $\mathbb{C}$-algebra, we know that there are $\mathbb{C}$-vector space isomorphisms

$$\begin{aligned}\text{Hom}_{\mathbb{C}}\left(\frac{\mathbb{C}[G]}{[\mathbb{C}[G],\mathbb{C}[G]]},\mathbb{C}\right) &= \{f\in\text{Hom}_{\mathbb{C}}(\mathbb{C}[G],\mathbb{C})\mid f(ghg^{-1}h^{-1})=0\text{ for all }g,h\in G\}\\ &= \{f\in\text{Hom}_{\textbf{Set}}(G,\mathbb{C})\mid f(ghg^{-1})=f(h)\text{ for all }g,h\in G\}\\ &= \text{Hom}_{\textbf{Set}}(\text{Cl}(G),\mathbb{C})\end{aligned}$$

By a previous result, the vector space has a basis given by the characters of G. We now define an inner product on the $\mathbb{C}$-space.

Definition 3.3.1 (The Inner Product Space of Class Functions)
Let G be a finite group. Define the inner product space of class functions of G to be the $\mathbb{C}$-vector space

$$\text{Hom}_{\textbf{Set}}(\text{Cl}(G),\mathbb{C})$$

together with the inner product given by

$$\langle f_1,f_2\rangle=\frac{1}{|G|}\sum_{g\in G}f_1(g)\overline{f_2(g)}$$

for $f_1,f_2\in\text{Hom}_{\textbf{Set}}(\text{Cl}(G),\mathbb{C})$.

Proposition 3.3.2
Let G be a finite group. Let ϕ,ψ be characters of G. Then we have $\langle\phi,\psi\rangle\in\mathbb{R}$.

Proof
Let ϕ,ψ be characters. Then we have

$$\begin{aligned}\overline{\langle\phi,\psi\rangle} &= \overline{\frac{1}{|G|}\sum_{g\in G}\phi(g)\overline{\psi(g)}}\\ &= \frac{1}{|G|}\sum_{g\in G}\overline{\phi(g)}\psi(g)\\ &= \frac{1}{|G|}\sum_{g\in G}\phi(g^{-1})\psi(g)\\ &= \frac{1}{|G|}\sum_{h\in G}\phi(h)\psi(h^{-1}) && (\text{Let } h=g^{-1})\\ &= \langle\phi,\psi\rangle\end{aligned}$$

so we are done. ■

Lemma 3.3.3
Let G be a finite group. Let V,W be irreducible $\mathbb{C}[G]$-modules whose representation is given by ρ and σ respectively. Let $f:V\to W$ be a morphism. Let $\widetilde{f}:V\to W$ be the map defined by

$$\widetilde{f}(v)=\frac{1}{|G|}\sum_{g\in G}\rho(g^{-1})\circ f\circ\sigma(g)$$

Then $\widetilde{f}$ is a $\mathbb{C}[G]$-module homomorphism. Moreover, we have

$$\widetilde{f}=\begin{cases}\frac{\text{tr}(f)}{\dim(V)}\text{id}_V & \text{if } V\cong W\\ 0 & \text{otherwise}\end{cases}$$

Proposition 3.3.4
Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$-modules. Then we have

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Theorem 3.3.5
Let G be a finite group. Let V, W be $\mathbb{C}[G]$-modules such that W is irreducible. Then we have

$$\langle \chi_V, \chi_W \rangle = \text{mult}_W(V)$$

Corollary 3.3.6
Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$-modules. Then $V \cong W$ if and only if $\chi_V = \chi_W$.

Proposition 3.3.7
Let G be a finite group. Let V be a $\mathbb{C}[G]$-module. Then V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Lemma 3.3.8
Let G be a finite group. Then we have

$$\dim_{\mathbb{C}}(\text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})) = |\text{Cl}(G)|$$

Lemma 3.3.9
Let G be a finite group. Let $\rho : G \to \text{End}_{\mathbb{C}}(V)$ be an irreducible representation of G. Let $f \in \text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$. Let $\varphi : V \to V$ be the map defined by

$$\varphi(v) = \sum_{g \in G} f(g)\rho(g)$$

Then we have

$$\varphi(v) = \frac{|G|}{\dim(V)} \langle f, \overline{\chi_V} \rangle v$$

for all $v \in V$.

Lemma 3.3.10
Let G be a finite group. Let $f \in \text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$-modules. If $\langle f, \chi_{V_i} \rangle$ for all $1 \leq i \leq n$, then we have $f = 0$.

Proposition 3.3.11
Let G be a finite group. Let $V_1, \dots, V_n$ be a complete list of non-isomorphic irreducible $\mathbb{C}[G]$-modules. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is an orthonormal basis for $\text{Hom}_{\textbf{Set}}(\text{Cl}(G), \mathbb{C})$.

Corollary 3.3.12
Let G be a finite group. Then there are exactly $|\text{Cl}(G)|$ non-isomorphic irreducible $\mathbb{C}[G]$-modules.

Corollary 3.3.13
Let G be a finite group. Let $g, h \in G$. Then g and h are conjugate if and only if $\chi(g) = \chi(h)$ for all characters χ of G.

Corollary 3.3.14
Let G be a finite group. Let $g \in G$. Then g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ of G.

3.4 Character Tables

Definition 3.4.1 (The Character Table of a Finite Group)
Let G be a finite group. Let $V_1, \dots, V_{|\mathrm{Cl}(G)|}$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$-modules. Let $\mathrm{Cl}(g_1), \dots, \mathrm{Cl}(g_{|\mathrm{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G. Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\mathrm{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \chi_{V_i}(g_j)$$

Example 3.4.2
The character table for $\mathbb{Z}/3\mathbb{Z}$ is given by

$\mathbb{Z}/3\mathbb{Z}$	$[0]$	$[1]$	$[2]$
χ_{Trivial}	1	1	1
χ_2	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_3	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$

where χ_2 is the character associated to the representation $\mathbb{Z}/3\mathbb{Z} \to \mathbb{C}$ given by $[1] \mapsto e^{2\pi i/3}$ and χ_3 is the character associated to the representation $\mathbb{Z}/3\mathbb{Z} \to \mathbb{C}$ given by $[1] \mapsto e^{4\pi i/3}$.

Example 3.4.3
The character table for $D_6 = \langle r, s \mid r^3, s^2, rsrs \rangle$ is given by

D_6	1	r	s
χ_{Trivial}	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

where χ_2 is the character associated to the representation $D_6 \to \mathbb{C}$ given by $r \mapsto 1$, $s \mapsto -1$ and χ_3 is the character associated to the representation

Example 3.4.4
The character table for A_4 is given by

A_4	1	$(1\ 2)(3\ 4)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
χ_{Trivial}	1	1	1	1
χ_2	1	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_3	1	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$
χ_3	3	-1	0	0

where χ_2 is the character associated to the representation $D_6 \to \mathbb{C}$ given by $r \mapsto 1$, $s \mapsto -1$ and χ_3 is the character associated to the representation

Proposition 3.4.5
Let G be a finite group. Let $A = (a_{i,j}) \in M_{|\mathrm{Cl}(G)|}(\mathbb{C})$ be the character table of G. Then the following are true.

- The rows of A are orthogonal:
$$\sum_{j=1}^{|\mathrm{Cl}(G)|} \frac{a_{r,j}\overline{a_{s,j}}}{|C_G(g_j)|} = \delta_{r,s}$$
- The columns of A are orthogonal:
$$\sum_{i=1}^{|\mathrm{Cl}(G)|} \frac{a_{i,r}\overline{a_{i,s}}}{|C_G(g_s)|} = \delta_{r,s}$$

Motivated by the above proposition, we define an augmented version of the character table as follows.

Definition 3.4.6 (The Augmented Character Table)
Let G be a finite group. Let $V_1, \dots, V_{|\mathrm{Cl}(G)|}$ be the complete list of non-isomorphic irreducible

$\mathbb{C}[G]$-modules. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G. Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \frac{\chi_{V_i}(g_j)}{|C_G(g_j)|}$$

Corollary 3.4.7
Let G be a finite group. Let A be the augmented character table of G. Then A is an orthonormal matrix.

A number of previous theorems help determining the character table. To compute the number of conjugacy classes, we may use the following facts. Suppose that $V_1, \dots, V_n$ is the complete list of non-isomorphic irreducible representations of a finite group G.

- Use the fact that $|G| = \sum_{i=1}^n \dim(V_i)^2$ and arithmetic to figure out n.
- $|\text{Cl}(G)| = n$.

To then figure out the character table, we may use the following facts.

- $\chi_{V_i}(1_G) = \dim(V_i)$.
- The trivial map $G \to \mathbb{C}$ is always a representation of G.
- Orthogonality relations between the rows and the columns
- $\chi(g^{-1}) = \overline{\chi(g)}$ for all characters χ.
- g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ.
- If χ is an irreducible character, then $\overline{\chi}$ is also n irreducible character.

Proposition 3.4.8
Let G be a finite group. Let χ be a character of G. Let $\rho : G \to \mathbb{C}^\times$ be a representation of G. Then the following are true.
- $\chi\rho$ is a representation of G.
- If χ is irreducible, then $\chi\rho$ is irreducible.

3.5 Representation Theory of Subgroups

Definition 3.5.1 (Restriction of Representation to a Subgroup)
Let k be a field. Let G be a group. Let $H \leq G$ be a subgroup. Let V be a representation of G over k.
- Define the restriction of V to H to be the vector space
$$\text{Res}_H^G(V) = V$$
together with the action of $k[H]$ given by restricting the action of $k[G]$ to $k[H]$.
- Define the restriction of χ_V to H by $\text{Res}_H^G \chi_V$.

Note: Even if V is an irreducible $k[G]$-module, it may not be an irreducible $k[H]$-module.

Proposition 3.5.2
Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a non-zero character of H. Then there exists an irreducible character χ of G such that $\langle \text{Res}_H^G \chi, \psi \rangle \neq 0$.

Proof
Let $\chi_1, \dots, \chi_k$ be the complete list of irreducible characters of G. Let χ_{reg} be the character of the regular representation of G. Then we know that $\chi_{\text{reg}} = \sum_{i=1}^k \chi_i(1_G)\chi_i$. Notice that the regular $\mathbb{C}[G]$-module contains $\mathbb{C}[H]$ and $\mathbb{C}[H]$ is closed under the action of H. Hence $\mathbb{C}[H]$ is a submodule of $\text{Res}_H^G(\mathbb{C}[G])$.

Suppose that ψ is irreducible. Then the corresponding $\mathbb{C}[H]$-module V is a submodule of the regular representation $\mathbb{C}[H]$. Thus $\langle \mathrm{Res}_H^G \chi_{\text{reg}}, \psi\rangle \neq 0$. Then we have

$$0 \neq \left\langle \sum_{i=1}^k \chi_i(1_G)\mathrm{Res}_H^G\chi_i, \psi \right\rangle = \sum_{i=1}^k \chi_i(1_G)\langle \mathrm{Res}_H^G\chi_i, \psi\rangle$$

Thus there exists χ_i such that $\langle \mathrm{Res}_H^G\chi_i, \psi\rangle \neq 0$.

We can extend this to the case where ψ is reducible by noting that for a character ψ_j for some irreducible composition factor of ψ, we have $\langle \mathrm{Res}_H^G\chi_i, \psi_j\rangle \neq 0$. ■

Theorem 3.5.3 (Clifford's Theorem)
Let G be a finite group. Let $H \trianglelefteq G$ be a normal subgroup. Let χ be an irreducible character of G. Let $\psi_1, \dots, \psi_n$ be the complete list of irreducible characters of H such that $\mathrm{Res}_H^G\chi$ is a linear combination of $\psi_1, \dots, \psi_n$ with non-zero coefficients. Then the following are true.

- Each ψ_i have the same dimension.
- $\langle \mathrm{Res}_H^G\chi, \psi_i\rangle$ is the same for all i.

Proof
Let V be an irreducible $\mathbb{C}[G]$-module corresponding to χ. Let U be an irreducible $\mathbb{C}[H]$-module of $\mathrm{Res}_H^G(V)$. For all $g \in G$, the set $g \cdot U$ is a subspace of V. For all $h \in H$, we have

$$h(gu) = g(g^{-1}hg)u \in gU$$

since U is a $\mathbb{C}[H]$-module and H is normal implies $g^{-1}hg \in H$. So $g \cdot U$ is a $\mathbb{C}[H]$-submodule of $\mathrm{Res}_H^G(V)$.

I claim that $g \cdot U$ is irreducible. Suppose that W is a $\mathbb{C}[H]$-submodule of $g \cdot U$. Then $g^{-1} \cdot W$ is a $\mathbb{C}[H]$-submodule of U by the above paragraph. Hence $g^{-1} \cdot W$ is U or 0 since U is irreducible. Hence W is equal to $g \cdot U$ or 0. Thus $g \cdot U$ is irreducible. Since $g \cdot -$ is an invertible linear map, we have $\dim_{\mathbb{C}}(g \cdot U) = \dim_{\mathbb{C}}(U)$.

Let $g_1, \dots, g_l$ be complete list of distinct elements in G such that $g_1 \cdot U, \dots, g_l \cdot U$ are pairwise not equal to each other. Then $\bigoplus_{j=1}^l g_j \cdot U$ is a $\mathbb{C}[G]$-submodule of V. Since V is an irreducible $\mathbb{C}[G]$-module, we have $\bigoplus_{g\in G} g \cdot U = V$. Hence V is the direct sum of these irreducible submodules of equal dimension. This proves part 1 of the theorem. ■

Proposition 3.5.4
Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be an irreducible character of G. Let $\psi_1, \dots, \psi_r$ be the complete list of irreducible characters of H. Suppose that $\mathrm{Res}_H^G\chi = \sum_{i=1}^r d_i\psi_i$. Then the following are true.

- We have $\sum_{i=1}^r d_i^2 \leq [G : H]$.
- The above equality holds if and only if $\chi(g) = 0$ for all $g \in G \setminus H$.

Proof
Notice that

$$\sum_{i=1}^r d_i^2 = \langle \mathrm{Res}_H^G\chi, \mathrm{Res}_H^G\chi\rangle_H = \frac{1}{|H|}\sum_{h\in H}\chi(h)\overline{\chi(h)}$$

and also

$$1 = \langle \chi, \chi\rangle_G = \frac{1}{|G|}\left(\sum_{h\in H}\chi(h)\overline{\chi(h)} + \sum_{g\in G\setminus H}\chi(g)\overline{\chi(g)}\right) = \frac{|H|}{|G|}\left(\sum_{i=1}^r d_i^2\right) + \frac{1}{|G|}\sum_{g\in G\setminus H}\chi(g)\overline{\chi(g)}$$

so we are done. ■

Definition 3.5.5 (The Induced Representation)
Let G be a finite group. Let $H \leq G$ be a subgroup. Let $g_1H, \dots, g_nH$ be the complete list of distinct left cosets of H in G. Let U be a $\mathbb{C}[H]$-module.

- Define the induced representation of U to be
$$\text{Ind}_H^G(U) = \bigoplus_{i=1}^n g_iU$$
together with the action of G given by left multiplication.
- Denote the character of the induced representation of χ by $\text{Ind}_H^G\chi$.

Example 3.5.6
Let $G = D_3 = \langle a, b \mid a^3, b^2, abab \rangle = S_3$. Let $H = \langle a \rangle \leq D_3$. Then the complete list of irreducible $\mathbb{C}[H]$-modules are given by

- $W_1 = \mathbb{C}\langle 1_H + a + a^2 \rangle$.
- $W_2 = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2 \rangle$.
- $W_3 = \mathbb{C}\langle 1_H + \omega^2 a + \omega a^2 \rangle$.

where $\omega = e^{2\pi i/3}$. Then the induced representations of the modules are given by

- $\text{Ind}_H^G(W_1) = \mathbb{C}\langle 1_H + a + a^2, b + ba + ba^2 \rangle$.
- $\text{Ind}_H^G(W_2) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.
- $\text{Ind}_H^G(W_3) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.

Proposition 3.5.7
Let G be a finite group. Let $H \leq G$ be a subgroup. Then there is an adjunction
$$\text{Ind}_H^G : \mathbf{Rep}_H \overset{\dashv}{\leftrightarrows} \mathbf{Rep}_G : \text{Res}_H^G$$

Theorem 3.5.8 (Frobenius Reciprocity Theorem)
Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be a character of G. Let ψ be a character of H. Then we have
$$\langle \text{Ind}_H^G\psi, \chi \rangle_G = \langle \psi, \text{Res}_H^G\chi \rangle_H$$

Proposition 3.5.9 (The Mackey Formula)
Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a character of H. Then we have
$$\text{Ind}_H^G\psi(g) = \frac{1}{|H|} \sum_{y \in G, ygy^{-1} \in H} \psi(ygy^{-1})$$

4 Representation Theory of the Symmetric Group

4.1

The conjugacy classes of S_n is the given by the cycle type of the permutations. These are indexed by partitions $\lambda = (\lambda_1, \dots, \lambda_k)$ such that $\sum_{i=1}^k \lambda_i = n$ and $\lambda_i \geq \lambda_{i+1}$. We write $\lambda \vdash n$. These can be drawn into Young diagrams:

- There are λ_i boxes in row i.
- Rows are left justified.

Let S_λ be the subgroup of S_n given by

$$S_\lambda = S_{1,\dots,\lambda_1} \times S_{\lambda_1+1,\dots,\lambda_1+\lambda_2} \times \cdots \times S_{n-\lambda_k+1,\dots,n} \cong S_{\lambda_1} \times \cdots \times S_{\lambda_k}$$

Let M^λ be the module associated to the character $\text{Triv}_{S_\lambda}^{S_n}$. This is not generally irreducible. But we can order partitions of n such that the following are true.

- $M^{\lambda(1)}$ is irreducible (call it $S^{\lambda(1)}$)
- $M^{\lambda(2)}$ contain copies of $S^{\lambda(1)}$ and a unique copy of a new irreducible module $S^{\lambda(2)}$.
- etc...

$S^{\lambda(i)}$ are called Specht modules. $M^{\lambda(i)}$ are called the permutation modules corresponding to λ. The ordering needed is reverse-lexicographic (revlex). Largest number to smallest number, starting with the first component and moving to latter components to break ties.

Let $\overline{t_i}$ denote the distinct λ-tabloids: fill the young diagram of λ with $\{1, \dots, n\}$ (this is called a Young tableau) and mark two as equivalent if each corresponding row has the same elements. Then $M^\lambda = \mathbb{C}\langle \overline{t_1}, \dots, \overline{t_k} \rangle$.

Example 4.1.1
The partitions of $\{1, \dots, 4\}$ is given by

$$\{(4), (3,1), (2,2), (2,1,1), (1,1,1,1)\}$$

Example 4.1.2
Let $\lambda = (n)$. Then $M^\lambda \cong \mathbb{C}$ and S_n acts on this trivially. Thus $\text{Ind}_{S_\lambda}^{S_n}\text{Triv} = \text{Triv}$.

Let $\lambda = (1, \dots, 1)$. Then $M^\lambda \cong \mathbb{C}^{n!}$. Then $\text{Ind}_{S_\lambda}^{S_n}\text{Triv} = \chi_{\text{reg}}$.

Let $\lambda = (n-1, 1)$. Then $M^\lambda \cong \mathbb{C}^n$. Then $\text{Ind}_{S_\lambda}^{S_n}\text{Triv}$ is the character of the permutation representation.

Definition 4.1.3 (Young Diagram)
Let $n \in \mathbb{N}^+$. Let $\lambda = (\lambda_1, \dots, \lambda_k)$ be a partition of n. Define the Young diagram of λ to be

$$Y(\lambda) = \{(i,j) \in \mathbb{N}^+ \times \mathbb{N}^+ \mid 1 \leq i \leq k, 1 \leq j \leq \lambda_i\}$$

Definition 4.1.4 (Young Tableau)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n. A Young tableau of λ is a bijection $t : Y(\lambda) \to \{1, \dots, n\}$. Denote the set of all Young tableau of λ by Δ^λ.

Definition 4.1.5 (Young Tabloid)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n.

- Let t_1, t_2 be Young tableau of λ. We say that t_1 and t_2 are equivalent if

$$\{t_1(r_0, j) \mid 1 \leq j \leq \lambda_i\} = \{t_2(r_0, j) \mid 1 \leq j \leq \lambda_k\}$$

for all $1 \leq r_0 \leq k$ (in other words the rows of the tableau are equal up to permutation).

- A Young tabloid of λ is an equivalence class $[t]$ where t is a Young tableau of λ.

Definition 4.1.6 (The Associated Polytabloid of a Tableau)
Let $n \in \mathbb{N}^+$. Let λ be a partition of n. Let t be a λ-tableau. Define the associated polytabloid to be

$$e(t) = \sum_{\sigma \in C(t)} \text{sign}(\sigma)(\sigma \cdot [t])$$

where $C(t) = \prod_{i=1}^k \text{Sym}(C_i)$ where C_i is the ith column of t.

Definition 4.1.7 (The Specht Module of a Partition)
Let $n \in \mathbb{N}^+$. Let λ be a partition for n. Define the Specht module associated to λ to be the $\mathbb{C}$-vector space

$$S^\lambda = \mathbb{C}\langle e(t) \mid t \text{ is a } \lambda\text{-tableau } \rangle$$

Theorem 4.1.8
Let $n \in \mathbb{N}^+$. Then $\{S^\lambda \mid \lambda \text{ is a partition for } n\}$ forms the complete list of non-isomorphic irreducible $\mathbb{C}[S_n]$-modules.

5 Some Combinatorics

5.1 Partitions

(To be separated to notes on combinatorics:)

Definition 5.1.1 (Partitions of an Integer) Let $n \in \mathbb{N}$. A partition λ of n is a list $(\lambda_1, \lambda_2, \dots, \lambda_l)$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_l > 0$ and $\sum_{i=1}^{l} \lambda_i = n$.

Definition 5.1.2 (Dominance of Partitions) Let λ and μ be two partitions of n. We say that λ dominates μ if for all $k \in \mathbb{N}$,

$$\sum_{i=1}^{k} \lambda_i \geq \sum_{i=1}^{k} \mu_i$$

In this case we denote it as $\lambda \trianglerighteq \mu$.

Lemma 5.1.3 Let $n \in \mathbb{N}$. Then the set of partitions of n together with dominance forms a poset.

Definition 5.1.4 (Lexicographical Ordering) Let λ and μ be partitions of n. We say that $\lambda < \mu$ if for some $i \in \mathbb{N}$, the following are true.

- For all $j < i$, $\lambda_j = \mu_i$
- $\lambda_i < \mu_i$

Lemma 5.1.5 Let $n \in \mathbb{N}$. Then the lexicographical order on the set of all partitions of n form a total ordering.

Proposition 5.1.6 Let λ and μ be partitions of n. If $\lambda \trianglerighteq \mu$, then $\lambda \geq \mu$.

5.2 Young Diagrams and Young Tabloids

Proposition 5.2.1 Let λ be a partition of n. Let T be the set of all λ-tabloids. Then S_n acts on T by

$$\tau \cdot [t_j] = [\tau(t_j)]$$

Definition 5.2.2 (Young Subgroup) Let λ be a partition of n. Define the Young subgroup of $S_\lambda \leq S_n$ to λ is defined as

$$S_\lambda = S_{1,\dots,\lambda_1} \times S_{\lambda_1+1,\dots,\lambda_1+\lambda_2} \times \cdots \times S_{n-\lambda_l+1,\dots,n}$$

Definition 5.2.3 (Row and Column Stabilizers) Let t be a tableau with rows $R_1, \dots, R_l$ and columns $C_1, \dots, C_k$. Define the row stabilizer of t to be the group

$$R_t = S_{R_1} \times \cdots \times S_{R_l}$$

Define the column stabilizer of t to be the group

$$C_t = S_{C_1} \times \cdots \times S_{C_k}$$

For $\tau \in R_t$, we have that $\tau([t]) = [t]$. In fact every element of $[t]$ is given in this way. Thus by writing

$$R_t t = \{\tau(t) \mid \tau \in R_t\}$$

we can identify the tabloid as

$$[t] = R_t t$$

6 Representations of the Symmetric Group

6.1 Representations Associated to a Partition

Recall that a partition $\lambda = (\lambda_1, \dots, \lambda_k)$ of an integer n gives a Young diagram, which n left-aligned boxes with λ_i boxes on row i. By writing $1, \dots, n$ in the boxes, we obtain a Young tableau. We then consider its equivalence class of Young tableau, in which we say that two Young tableau are equivalent if row i of the two tableau contains the same element for all i. Such an equivalence class is called a Young tabloid of shape λ.

We have also seen that S_n acts on the set of all tabloids by permuting the numbers in the boxes. The stabilizer of a tabloid t is precisely the row stabilizers $R_t = S_{R_1} \times \cdots \times S_{R_l}$, for $R_1, \dots, R_l$ the rows of the tabloid.

Definition 6.1.1 (Permutation Module Associated to a Partition) Let λ be a partition of n. Let $[t_1], \dots, [t_k]$ be a complete list of λ-tabloids. Define the permutation module associated to λ to be

$$M^\lambda = \mathbb{C}\langle [t_1], \dots, [t_k] \rangle$$

This means that $\rho : S_n \to \text{GL}(M^\lambda)$ where

$$\rho(\tau) : M^\lambda \to M^\lambda$$

for $\tau \in S_n$ is defined on the basis by sending $[t_j]$ to $[\tau(t_j)]$.

Proposition 6.1.2 Let λ be a partition of n. Let $T = S_n / S_\lambda$ be the set of cosets of S_λ. Then the following representations are equivalent.

- The induced representation $\text{Ind}_{S_\lambda}^{S_n} 1$
- The permutation representation $\rho : S_n \to \text{GL}(\mathbb{C}T)$ where $\rho(\tau) : \mathbb{C}T \to \mathbb{C}T$ for $\tau \in S_n$ is defined on the basis by sending $\pi_k S_\lambda$ to $\tau \pi_k S_\lambda$
- The permutation module $\psi : S_n \to \text{GL}(M^\lambda)$ corresponding to λ

Recall that for a tableau t, the columns stabilizer of t is given by

$$C_t = S_{C_1} \times \cdots \times S_{C_k}$$

if $C_1, \dots, C_k$ are the columns of the tableau.

Definition 6.1.3 (Associated Polytabloid) Let λ be a partition of n. Let t be a λ-tableau. Define the associated polytabloid $e_t \in M^\lambda$ by

$$e_t = \sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [t])$$

Lemma 6.1.4 Let t be a tableau of n boxes and let $\pi \in S_n$ be a permutation. Then the following are true.

- $R_{\pi t} = \pi R_t \pi^{-1}$
- $C_{\pi t} = \pi C_t \pi^{-1}$
- $\kappa_{\pi t} = \pi \kappa_t \pi^{-1}$
- $e_{\pi t} = \pi e_t$.

6.2 Representations of the Symmetric Group

Definition 6.2.1 (Specht Module) Let λ be a partition of n. Define the Specht module S^λ to be the submodule of M^λ where

$$S^\lambda = \mathbb{C}\langle e_t \mid t \text{ has shape } \lambda \rangle$$

Lemma 6.2.2 Let $n \in \mathbb{N}$. Then the following are true

- Then the representation $\rho : S_n \to \text{GL}(S^{(n)})$ is the trivial representation.
- The representation $\rho : S_n \to \text{GL}(S^{(1^n)})$ is equivalent to the sign representation.

Lemma 6.2.3 Let λ and μ be partitions of n. Let t be a λ-tableau and let s be a μ-tableau. Then the following are true.

- If $\sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [s]) \neq 0$, then $\lambda \trianglerighteq \mu$.
- If $\lambda = \mu$, then $\sum_{\pi \in C_t} \text{sign}(\pi)(\pi \cdot [s]) \in \{-e_t, e_t\}$.

Theorem 6.2.4 (The Submodule Theorem) Let λ be a partition of n. Let U be a submodule of M^λ. Then either $S^\lambda \subseteq U$ or $U \subseteq (S^\lambda)^\perp$.

Lemma 6.2.5 Let λ and μ be partitions of n. Let $f \in \text{Hom}_{S_n}(S^\lambda, M^\mu)$ be non-zero. Then $\lambda \trianglerighteq \mu$. If $\lambda = \mu$ then f is multiplication by a scalar.

Proof S^λ is generated by e_t by definition. Since $f \neq 0$, there exists a tableau t such that $f(e_t) \neq 0$. Since S^λ is a submodule of M^λ, we have that $M^\lambda = S^\lambda \oplus (S^\lambda)^\perp$. Define an extension $\tilde{f} : M^\lambda \to M^\mu$ of f as follows. For $v \in (S^\lambda)^\perp$, set $\tilde{f}(v) = 0$. Now we have that ??? (5.4.2) ■

Theorem 6.2.6 Let $n \in \mathbb{N}$. Then the set of all S^λ for λ a partition of n forms a complete list of irreducible S_n-representations.

Proof We first prove that S^λ is irreducible. Let $U \subseteq S^\lambda$ be a subrepresentation. By the submodule theorem, either $S^\lambda \subseteq U$ or $U \subseteq (S^\lambda)^\perp$. Thus $S^\lambda = U$ or $U \subseteq S^\lambda \cap (S^\lambda)^\perp = 0$.

It is clear that the number of conjugacy classes of S_n is equal to the number of partitions of n. Thus the set of all S^λ for λ a partition of n has cardinality equal to the the number of conjugacy classes of S_n. By theorem 2.4.6, the number of pairwise non-isomorphic irreducible representations of S_n is equal to the number of conjugacy classes of S_n. Thus it remains to show that they are pairwise non-isomorphic.

Suppose that S^λ is equivalent to S^μ for some partitions λ and μ of n. Then there exists a non-zero $f \in \text{Hom}_{S_n}(S^\lambda, S^\mu)$ which can be interpreted as an element of $\text{Hom}_{S_n}(S^\lambda, M^\mu)$. By the above lemma, we conclude that $\lambda \trianglerighteq \mu$. Similarly, we conclude that $\mu \trianglerighteq \lambda$. Thus $\lambda = \mu$. ■

6.3 Character Tables for S_3

Proposition 6.3.1 (The Sign Representation) Let $n \in \mathbb{N}$. Then the map $\rho : S_n \to \mathbb{C}^*$ defined by

$$\rho(\tau) = \text{sign}(\tau)$$

for $\tau \in S_n$ is a one dimensional complex representation of S_n.

Recall that any two permutations in S_n of the same cycle type are conjugate. This completely determines the conjugacy classes of S_n. In particular, this means that the number of conjugacy classes of S_n is given by the number of integer partitions of n.

Proposition 6.3.2 Let $X = \{a, b, c\}$ such that S_3 acts on X by permutation. Consider the permutation representation $\rho : S_3 \to \text{GL}(\mathbb{C}X)$. Then the subspace

$$U = \mathbb{C}\langle a + b + c\rangle$$

of $\mathbb{C}X$ is a subrepresentation of G.

Proof We want to show that $tau \cdot u \in U$ for all $u \in U$. We just have to check this for the basis. We have that

$$tau \cdot (a + b + c) = \tau(a) + \tau(b) + \tau(c)$$

Since τ is a permutation, $\{\tau(a), \tau(b), \tau(c)\} = \{a, b, c\}$ hence we are done. ■

Proposition 6.3.3 Let $X = \{a, b, c\}$ such that S_3 acts on X by permutation. Consider the permutation representation $\rho : S_3 \to \text{GL}(\mathbb{C}X)$ and the subrepresentation $U = \mathbb{C}\langle a + b + c\rangle$ of $\mathbb{C}X$. Then V/U is an irreducible representation of G.

Proof We already know that V/U is a representation, it remains to show that it is irreducible. We wish to invoke 2.3.5. The representation $\overline{\rho} : S_3 \to \text{GL}(V/U)$ is given by $\rho(\tau)(v+U) = \tau(v)+U$. We give a non-canonical basis of $\{a + V/U, b + V/U\}$ to V/U. The character of V/U is given as follows.

There are three conjugacy classes of S_3 given by $\{1\}, \{(a, b), (b, c), (a, c)\}$ and $\{(a, b, c), (a, c, b)\}$. We have that

- $\chi_{\overline{\rho}}(1) = \text{tr}(I) = 2$
- Under the given basis, the matrix of $\overline{\rho}(a, b)$ is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Hence $\chi_{\overline{\rho}}((a, b)) = 0$
- Under the given basis, the matrix of $\overline{\rho}(a, b, c)$ is $\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Hence $\chi_{\overline{\rho}}((a, b, c)) = -1$

Now we have that

$$\begin{aligned} \langle \chi_{\overline{\rho}}, \chi_{\overline{\rho}}\rangle &= \frac{1}{|S_3|} \sum_{\tau \in S_3} \chi_{\overline{\rho}}(\tau)^2 \\ &= \frac{1}{6}(1 \cdot 2^2 + 3 \cdot 0 + 2 \cdot (-1)^2) \\ &= 1 \end{aligned}$$

By lemma 2.3.5, we conclude that V/U is irreducible. ■

Theorem 6.3.4 There are a total of three irreducible complex representations of S_3 given by

- The trivial representation $\rho_1(\tau) = 1$
- The sign representation $\rho_2(\tau) = \text{sign}(\tau)$
- The representation $\overline{\rho}$ given by 6.3.3.

Proof We have seen that each of these are irreducible. There are three conjugacy classes of S_3. By 2.4.6 we conclude that we have exhausted all the irreducible representations. ■

Theorem 6.3.5 The character table of S_3 is given as follows.

G	1	$\{(1,2),(1,3),(2,3)\}$	$\{(1,2,3),(1,3,2)\}$
χ_{trivial}	1	1	1
χ_{sign}	1	-1	1
$\chi_{\overline{\rho}}$	2	0	-1

6.4 Representations of the Alternating Group